# Project Proposal

**on**

# AI Integrated Machine vision technology for equipment condition monitoring and feed control in Indian Industries with a use case from our region

**Submitted to**

## Hyundai Hope Scholarship

**By**


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# PART I: EXECUTIVE SUMMARY

**Project Title:** AI Integrated Machine vision technology for equipment condition monitoring and feed control in Indian Industries with a use case from our region

**Project Team Details:**

| Team leader | |
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# PART II: DETAILED PROPOSAL

**Problem Statement:**

Upon physical survey of various industries in the state of Chhattisgarh, we found that separation of mixed input materials in production line is a critical task in many industries, that directly impacts efficiency, productivity, and product quality. Traditional manual sorting processes are labor-intensive, prone to human error, and often inconsistent, leading to increased operational costs and decreased productivity. While some automated systems exist, they frequently lack the flexibility and precision required to handle diverse materials and adapt to varying industrial needs.

The challenge is further compounded by the need for rapid processing speeds to meet high production demands, while simultaneously ensuring accuracy in the separation process. This is particularly important in industries such as manufacturing, recycling, and agriculture, where material differentiation by attributes like size, shape, color, or material type is essential and at the same time, the exterior profiles are not standard shapes to use simple machine vision models.

**Our use case:**

Specifically, we took the recycling industry as a use case, where the input material is a bunch of wire that goes through a conveyor, which often chocks the processing unit. Here, the problem is not only to declutter the input material, but also to estimate when the input material is required to be decluttered and when it should be sent as it is. Further, the same material, due to its dis-continuous feed to the down-stream equipment, leads to frequent choking. There is a need to do condition monitoring for this equipment and develop an adaptive feed model to effectively utilize the machines.

Here, we are proposing a camera-based feed monitoring system that shall guide a robotic arm for decluttering and then the data from camera shall be used to control the feed to the input of mentioned equipment. While the solution through this project appears specific to this industry, the system and protocol we aim at developing shall serve a larger purpose for industries of this nature, and we feel many such industries exist across the country. Through this project, we aspire to establish a technology startup as well.

**Objective:**

1. **Robotic Arm Development:**
    - Design a robotic arm capable of handling a wide variety of materials with precision and speed.
    - Testing the arm on performance for decluttering tasks with high reliability.
2. **AI-based Machine Vision Strategy Development:**
    - Implement state-of-the-art computer vision algorithms to identify and classify materials based on specific attributes such as size, shape, color, and material type.

- Enable real-time processing through on-board data processing to ensure rapid decision making and accurate separation.

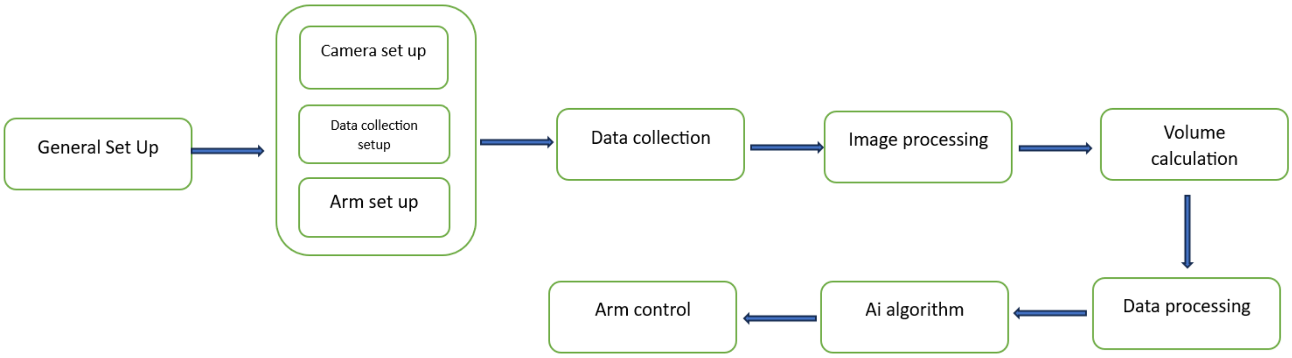
3. **Improving the process efficiency:**

    - To integrate other variables (such as voltage, current, and speed) to the developed model to create a robust decision-making model. Additionally, we aim to determine the optimal input ratio (e.g., voltage-to-current) that yields the desired results during machine operation, as required for the specific industry.

By addressing these objectives, the project aims to deliver a robust, flexible, and efficient solution that meets the evolving needs of modern industrial material handling and separation. This will not only improve productivity and reduce operational costs but also set a new standard for automation in industrial settings.

**Methodology:**

The overall project plan is presented as a flow chart below.




1. **Camera Setup and Data Collection**:

    - Set up the camera system in real production environment to capture relevant data (background data for filter, Trial data on raw materials, etc.).

    - Real-time images collection and setting up the system for onboard processing and cloud transfer/storage.

2. **Image Processing**:

    - Process the collected images to extract meaningful information:

        - Identify the raw material of interest (wires in this case).

        - volume estimation (e.g., geometric measurements, 3D reconstruction, etc.).

3. **Arm Control and Material Handling**:

- Integrate the image processing system with the developed arm (robotic or automated) to handle the material (decluttering).
- Based on the calculated volume and density, program the arm to pick and place the material efficiently into the machine

**Preliminary work:**

We have been working on this problem for the last three months since our summer break this year. We have spent some time understanding the entire process of the industry. Alongside that, we have also conducted extensive surveys to understand state-of-art in image processing and feature extraction algorithms with precision and speed.

We have procured LADER based camera for data capture. We have started working on algorithms for object recognition and volume estimation for our material (cluttered wire, here). We had conducted preliminary tests to calculate volume using YOLOv8, OpenCV, and Open3D. Here are the details of the tests and their outcomes:

*Segmentation of Irregular Objects:*

**Method: Used the SAM model for segmentation.**

*Outcome:*

The process took more than 5 minutes, making it impractical for our needs. Therefore, we decided to discontinue this approach.

*Depth Sensing Camera:*

**Method: Utilized a depth sensing camera to determine the distance between the camera and the object.**

*Outcome*:

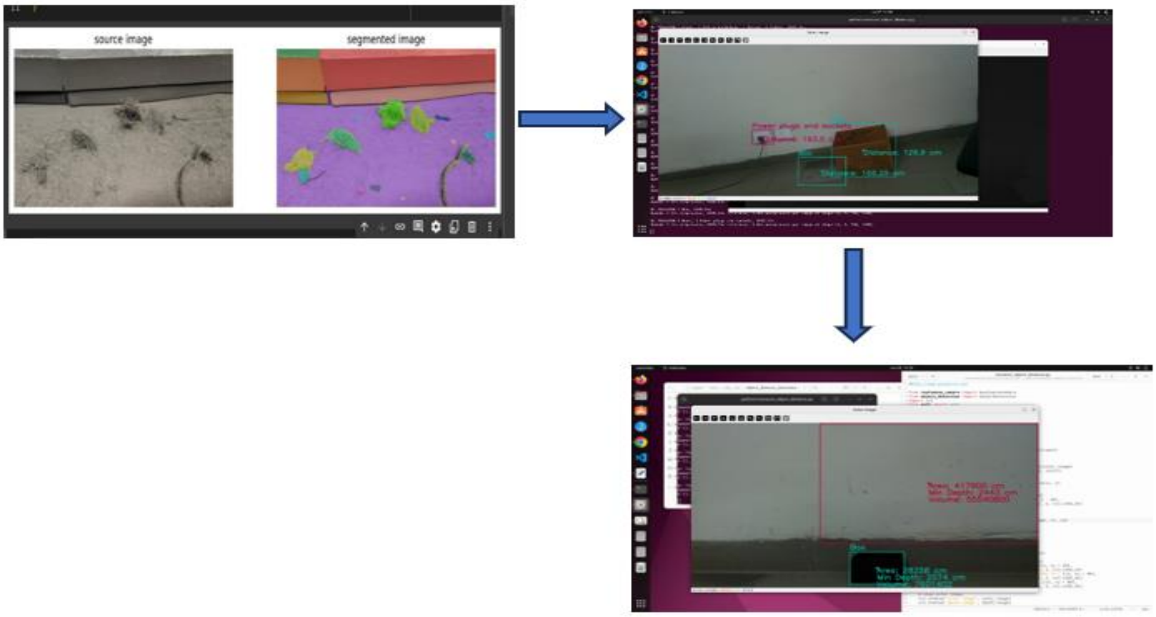
Successfully measured the distance, which is crucial for accurate volume calculation.

*Volume Calculation of Defined Objects:*

**Method: Calculated the volume of a defined object.**

*Outcome:*

Achieved approximately 55% accuracy. However, this method failed when applied to irregular objects.




**Budget estimate:**

We have already procured the components required for image capture and processing from our pocket. We need financial assistance for components required to build the robotic arm, including the actuators, and fabricators cost. Accordingly, the following is our budget requirement.

| Sl. No. | Budget head | Amount |
|---|---|---|
| 1. | Consumables | 45,000 |
| 2. | Contingent expenditure | 10,000 |
| 3. | Travel expenditure | 5,000 |
| 3. | Institute overhead | 12,000 |
| | **Total** | **72,000** |

## 1. Future scope:

Upon successful demonstration of our technology, we want to scale up our developed technology for implementation in other industries. We would also like to explore the possibilities of creating a technological service-based start-up. Along with the solution solved through our project, we would also like to work on the following problems in future.

1. **System for Uniform Material Separation:**
   - Design the system to perform consistent and uniform separation according to predefined criteria through mechatronic intervention.

- Ensure the system can adapt to new separation criteria with minimal reconfiguration through data-based training.

2. **Create a User-Friendly Interface:**
    - Design an intuitive interface that allows operators to configure and control the system easily.
    - Provide comprehensive training and support materials to ensure effective system operation and troubleshooting.

3. **Scaling up for other systems:**
    - Expand the technology to solve other similar problems.
    - Incorporation of other data acquisition systems to bring a multi-sensor approach for object detection and process planning.